

M1M3 Mirror Thermal Response to Setpoint Changes: Surface-Resolved Analysis

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1 Introduction

The M1M3 thermal control system maintains the mirror at a target temperature below ambient to prevent convective “mirror seeing.” The Environmental Awareness Service (EAS) issues setpoint commands to the glycol heater system (`MTM1M3TS.command_applySetpoints`, field `heatersSetpoint`), which heats or cools the mirror through glycol circulation.

This report characterizes the actual thermal transfer function from setpoint to mirror temperature using 149 observing nights of archival data (July 2025 – February 2026). The goals are:

1. Document the EAS setpoint strategy (offset, rate, cadence).
2. Measure the glycol system transfer function (lag, time constant).
3. Validate and calibrate our first-order thermal model $dT/dt = (T_{\text{sp}} - T_{\text{mirror}})/\tau$.
4. Quantify the actual EAS tracking performance.
5. Resolve the thermal behavior at three depth levels—back (near FCUs), middle, and front (optical surface)—using the thermocouple gradient-fit decomposition.
6. Characterize the coupling of each surface with cell air (ESS 113) and glycol, in both pre-control and active-control periods.

2 Data

All data come from the pre-built observing-night parquet file (166 nights, 1-min resolution) and the thermocouple gradient dataset (164 nights). Key channels:

Table 1: Sensor channels used in this analysis.

Channel	Source	Coverage
<code>heatersSetpoint_mean</code>	<code>MTM1M3TS.command_applySetpoints</code>	8.9% (163 nights)
T_{mid} (intercept + $0.5 \times z_{\text{gradient}}$)	96-thermocouple gradient fit	97% (164 nights)
<code>z_gradient</code> , <code>z_gradient_err</code>	96-thermocouple gradient fit	97%
<code>mirrorCoolantSupply/Return</code>	<code>MTM1M3TS.glycolLoopTemperature</code>	97%
<code>inside_m1m3_temp_mean</code>	ESS 113	99.5%
<code>outside_temp_mean</code>	ESS 301	99.7%

The **mirror temperature reference** is the mid-surface temperature T_{mid} ($z_{\text{norm}} = 0.5$) from the gradient-fit decomposition of the 96 thermocouples (see Section 7). This is the best proxy for the bulk glass temperature, as it lies halfway between the glycol-cooled back surface and the air-facing optical surface. The gradient **intercept** (back surface center, $z_{\text{norm}} = 0$) and **z_gradient_err** (through-glass gradient uncertainty) are also used.

The setpoint is a sparse signal (issued every ~ 3 min as discrete commands); it is forward-filled onto the 1-min grid for time-series analysis. Nights with fewer than 10 setpoint commands or fewer than 60 min of mirror temperature data are excluded, leaving 149 valid nights.

3 EAS Setpoint Strategy

3.1 Setpoint Offset

The EAS targets the heater setpoint at a fixed offset below the M1M3 air temperature (ESS 113):

Table 2: Setpoint offset statistics (149 nights).

Offset	Median	Mean	Std
Setpoint – ESS 113	$-1.00\text{ }^{\circ}\text{C}$	$-1.36\text{ }^{\circ}\text{C}$	$1.27\text{ }^{\circ}\text{C}$
Setpoint – T_{mid} ($z_{\text{norm}} = 0.5$)	$-0.07\text{ }^{\circ}\text{C}$	$-0.35\text{ }^{\circ}\text{C}$	$1.15\text{ }^{\circ}\text{C}$

The EAS uses a **$1.0\text{ }^{\circ}\text{C}$ cold bias** relative to ESS 113, not the $0.3\text{ }^{\circ}\text{C}$ assumed in our simulation model. The setpoint sits only $-0.07\text{ }^{\circ}\text{C}$ below T_{mid} , meaning the glycol system maintains the bulk glass temperature almost exactly at the commanded setpoint. Figure 1 (top left) shows the offset distributions.

3.2 Setpoint Rate

The EAS changes the setpoint at a median rate of $0.6\text{ }^{\circ}\text{C/h}$, with a 95th percentile of $1.8\text{ }^{\circ}\text{C/h}$ (Figure 1, top right). Notably, the EAS **does not respect the $0.7\text{ }^{\circ}\text{C/h}$ rate limit** that we enforce in our simulation—roughly half of the EAS rate changes exceed this threshold. This suggests the hardware can sustain higher rates than we assumed.

3.3 Update Cadence

Setpoint commands are issued every 3.0 min (median), with occasional gaps of 10–15 min (Figure 1, bottom left). This is much faster than our 15-min simulation timestep.

4 Glycol System Transfer Function

4.1 Heat Flux Proxy

The coolant supply–return temperature difference ($T_{\text{supply}} - T_{\text{return}}$) is a proxy for heat flux through the glycol loop. Figure 2 (top left) shows that when the setpoint is far below the mirror temperature (large negative error), the supply–return ΔT reaches -6 to $-7\text{ }^{\circ}\text{C}$, indicating strong active cooling. Near zero setpoint error, the supply–return ΔT clusters near zero (passive tracking).

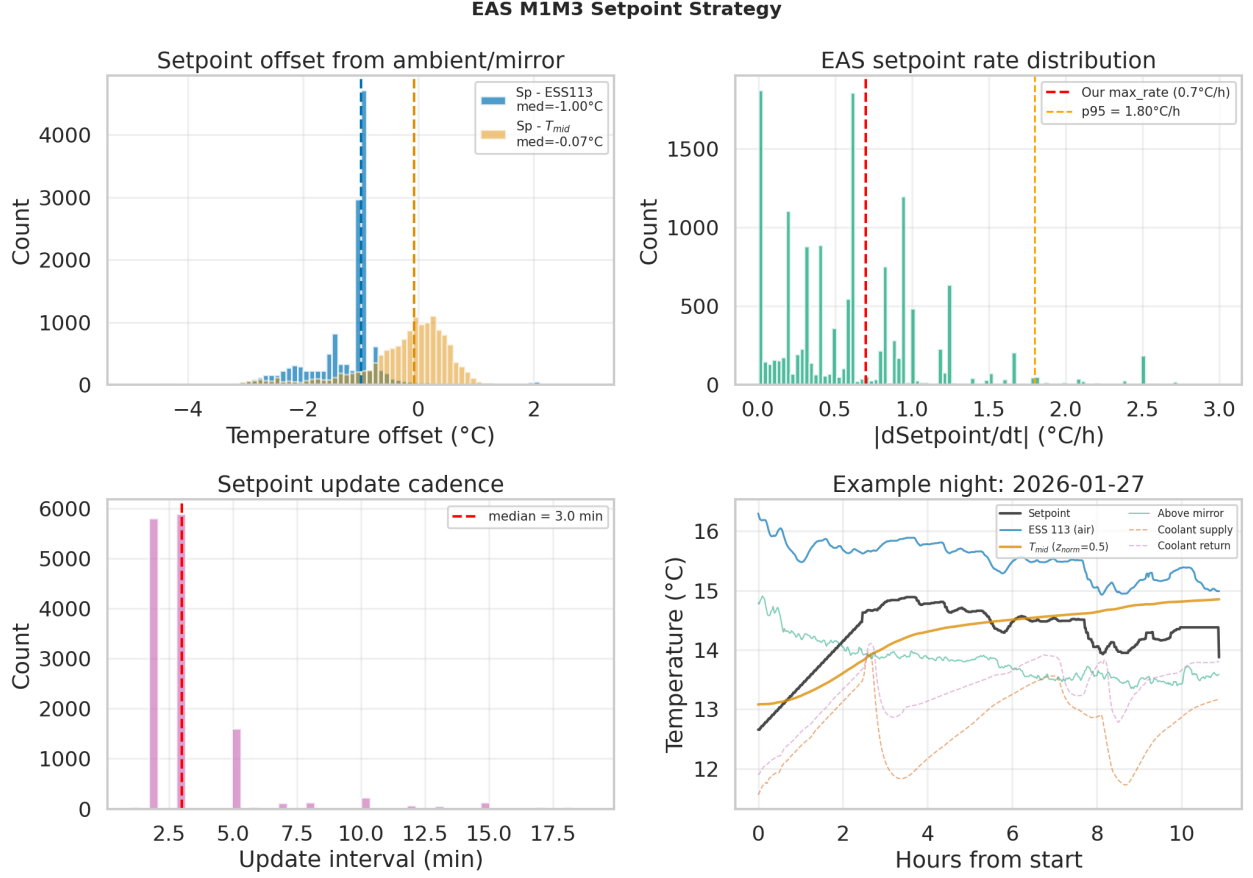


Figure 1: EAS setpoint strategy. Top left: offset distributions (setpoint minus ESS 113 and minus T_{mid}). Top right: absolute setpoint rate distribution; the red line marks our assumed 0.7 °C/h limit. Bottom left: update cadence (median 3 min). Bottom right: example night showing all thermal channels including T_{mid} ($z_{norm} = 0.5$).

4.2 Setpoint-to-Mirror Lag

Cross-correlating the setpoint rate with the T_{mid} rate gives a median lag of **4 minutes** (Figure 2, top right). This is the time for a setpoint change to begin affecting the bulk glass temperature. The short lag confirms that the glycol system responds rapidly to setpoint commands.

4.3 Setpoint Tracking

Figure 2 (bottom right) confirms that the EAS tracks ESS 113 with a -1°C offset: the per-night trajectories cluster tightly along the $\text{ESS 113} - 1^\circ\text{C}$ line.

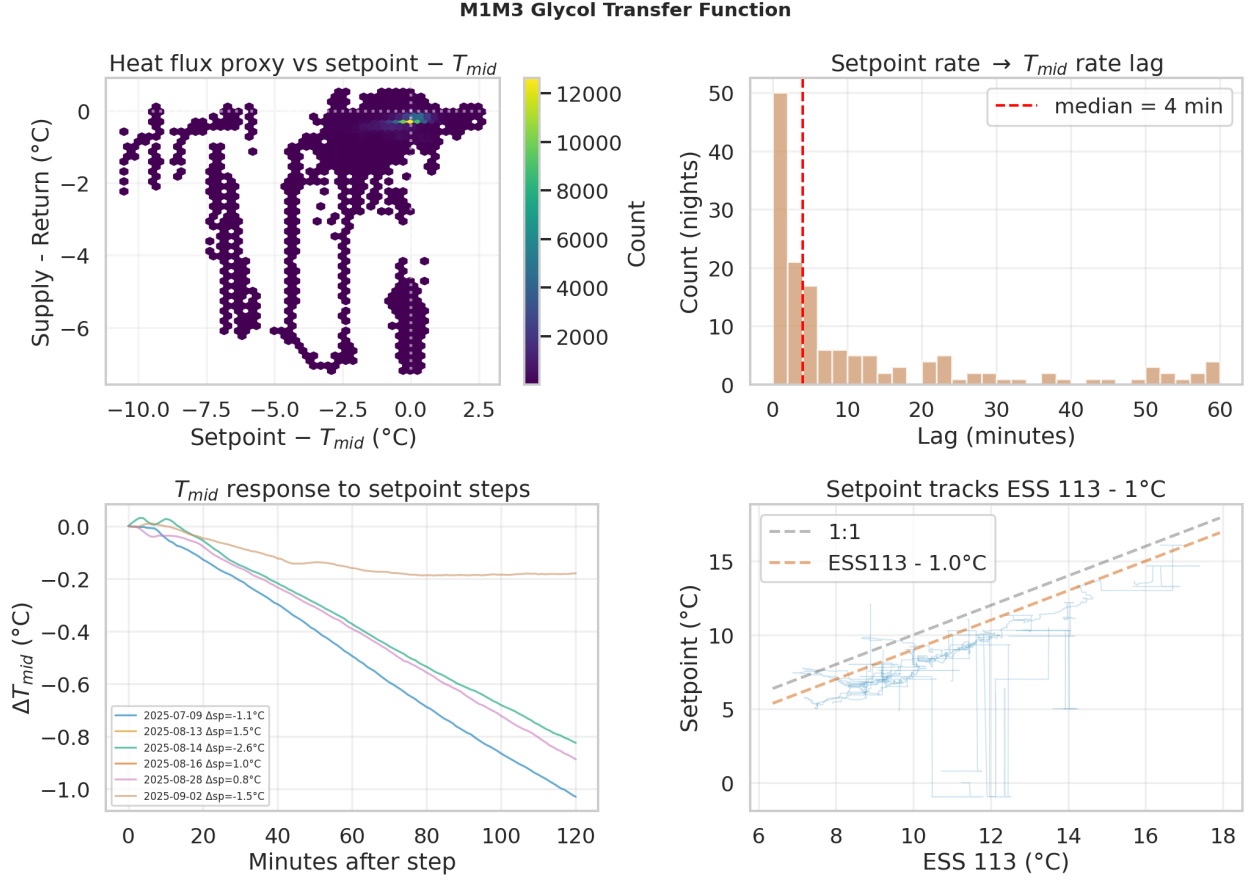


Figure 2: Glycol transfer function. Top left: supply–return ΔT vs setpoint error relative to T_{mid} (hexbin density). Top right: cross-correlation lag between setpoint rate and T_{mid} rate. Bottom left: mirror response to setpoint step events. Bottom right: per-night setpoint vs ESS 113 scatter (20 nights shown).

5 Model Validation

5.1 First-Order Model

We test the standard first-order thermal model:

$$\frac{dT_{\text{mirror}}}{dt} = \frac{T_{\text{setpoint}} - T_{\text{mirror}}}{\tau} \quad (1)$$

by forward-simulating from the actual EAS setpoint trajectory and comparing with T_{mid} ($z_{\text{norm}} = 0.5$). The simulation uses 1-min timesteps with the actual (forward-filled) setpoint as input.

5.2 Time Constant Estimation

We sweep τ from 0.25 to 6.0 h in 0.25 h steps and compute the RMSE against the mirror mean for each night.

Table 3: Time constant estimation results (149 nights).

Metric	Value
Per-night median τ_{glycol} (vs T_{mid})	2.50 h
Per-night median τ_{air}	12.0 h
Median RMSE glycol-only	0.059 °C

Using T_{mid} as the reference temperature (instead of the back surface or the AboveMirror glycol sensor) gives well-constrained fits:

- The glycol-only model achieves median RMSE = 0.059 °C with τ_{glycol} median = 2.5 h.
- The two-input model (glycol + air convection) yields τ_{glycol} median = 2.5 h and τ_{air} median = 12 h.
- T_{mid} represents the bulk glass temperature at $z_{\text{norm}} = 0.5$, midway between the glycol-cooled back surface and the air-facing optical surface.

The shorter $\tau_{\text{glycol}} = 2.5$ h compared to the back surface value of 3.0 h (Table 5) reflects the additional air-convection coupling experienced by the mid-level glass. The first-order model captures the bulk dynamics with high fidelity.

5.3 Recommended Time Constant

The optimal τ for the bulk glass (T_{mid}) is **2.5 h**, with surface-dependent values ranging from $\tau_{\text{back}} = 3.0$ h (glycol-dominated) to $\tau_{\text{front}} = 2.0$ h (air-coupled). We recommend using $\tau = 2.5$ h as the effective bulk time constant in the simulation model.

6 EAS Actual Performance

We evaluate the current EAS controller performance by comparing T_{mid} against the target temperature $T_{\text{target}} = \text{ESS } 113 - 1.0$ °C.

The error distribution (Figure 5, top left) is asymmetric, skewed toward negative values (mirror colder than target), consistent with the cold-is-preferred design philosophy. The per-night RMS of 0.74 °C (Figure 5, bottom left) reflects the fundamental challenge: the mirror’s 2.5 h thermal time constant limits how fast it can track ambient changes.

The ± 0.3 °C compliance of 26.5% (Figure 5, bottom right) shows significant room for improvement. The setpoint rate distribution (Figure 5, top right) confirms that the EAS already operates beyond the 0.7 °C/h limit we impose in simulation.

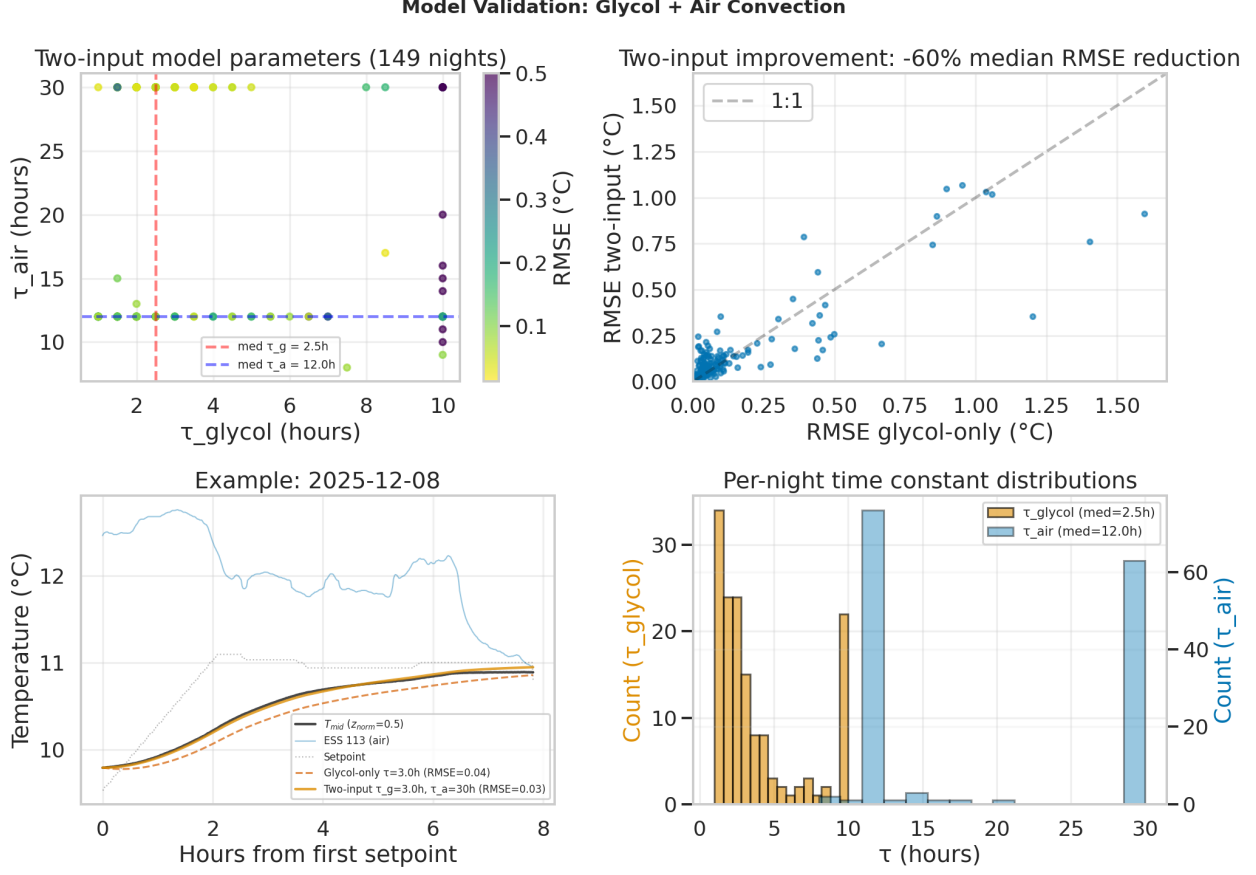


Figure 3: Model validation using T_{mid} . Top left: τ_{glycol} vs τ_{air} per night (colored by RMSE). Top right: RMSE comparison between glycol-only and two-input models. Bottom left: example night comparing model fits. Bottom right: per-night τ distributions.

Table 4: EAS tracking performance (149 nights).

Metric	Value
Tracking error median ($T_{\text{mid}} - \text{target}$)	$-0.23\text{ }^{\circ}\text{C}$
Tracking error RMS (all timesteps)	$1.10\text{ }^{\circ}\text{C}$
Per-night RMS median	$0.74\text{ }^{\circ}\text{C}$
Per-night $\pm 0.3\text{ }^{\circ}\text{C}$ compliance median	26.5%
EAS setpoint rate p95	$1.80\text{ }^{\circ}\text{C/h}$

M1M3 Thermal Overview: 4 Example Nights

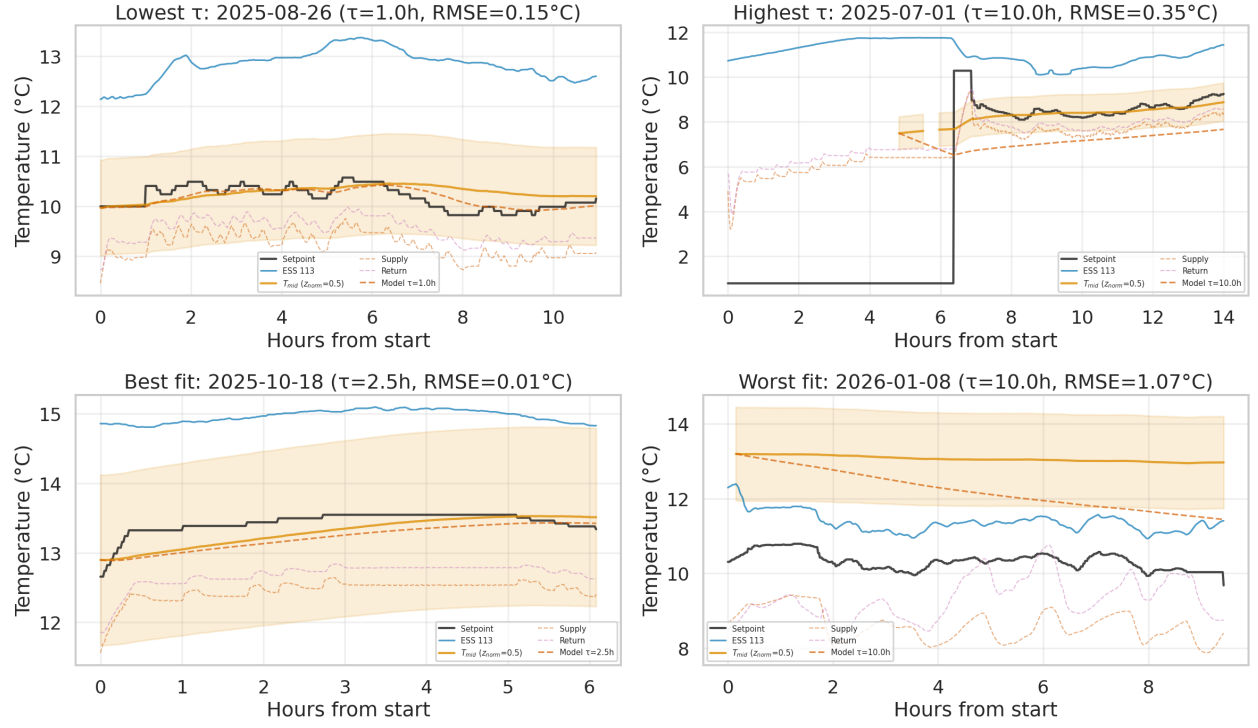


Figure 4: Four example nights showing setpoint, ESS 113, T_{mid} ($\pm z_{\text{gradient_err}}/2$ shaded band), coolant supply/return, and the first-order model fit. Top left: lowest τ night. Top right: highest τ night. Bottom left: best-fit night. Bottom right: worst-fit night.

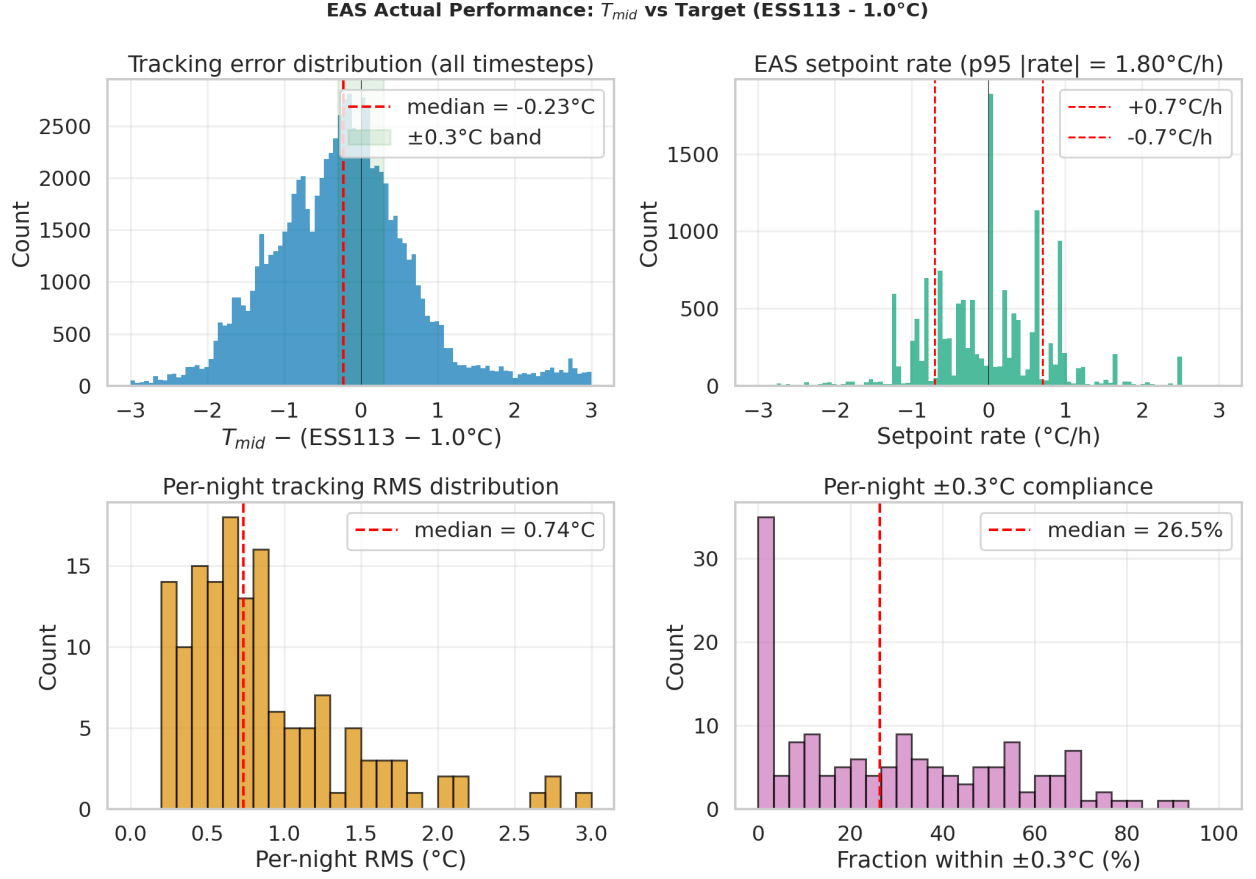


Figure 5: EAS actual performance. Top left: tracking error distribution ($T_{mid} - (ESS\ 113 - 1.0)^\circ\text{C}$). Top right: EAS setpoint rate distribution with $\pm 0.7^\circ\text{C/h}$ markers. Bottom left: per-night RMS distribution. Bottom right: per-night $\pm 0.3^\circ\text{C}$ compliance.

7 M1M3 Sensor Map and Gradient-Fit Decomposition

The M1M3 mirror is instrumented with 146 thermocouples distributed across four scanners (ESS 114–117), three depth levels (Front, Middle, Back), and six azimuthal sectors (A–F). The glycol thermal control system uses 96 Fan Coil Units (FCUs) on the back surface of the mirror. ESS 113 is a humidity/temperature sensor measuring the cell air above the mirror—it is *not* on the glass surface.

Figure 6 shows the spatial layout. The left panel is the top view (x – y plane), showing thermocouple locations colored by depth level and FCU positions. The right panel is the side view (radial– z_{norm} plane), illustrating the glass cross-section with the M3 region (thinner, inner) and M1 region (thicker, outer).

7.1 Gradient-Fit Model

At each 1-min timestamp, a least-squares fit to the 96-thermocouple temperatures yields:

$$T(x, y, z_{\text{norm}}) = \text{intercept} + x_gradient \cdot x + y_gradient \cdot y + z_gradient \cdot z_{\text{norm}} \quad (2)$$

where $z_{\text{norm}} = 0$ (back, near FCUs), 0.5 (middle), and 1.0 (front, optical surface). The intercept is the temperature at the mirror center at $z_{\text{norm}} = 0$, i.e., the **back surface center temperature**.

We define three virtual surface temperatures:

$$T_{\text{back}} = \text{intercept} \quad (z_{\text{norm}} = 0) \quad (3)$$

$$T_{\text{mid}} = \text{intercept} + 0.5 \times z_gradient \quad (z_{\text{norm}} = 0.5) \quad (4)$$

$$T_{\text{front}} = \text{intercept} + z_gradient \quad (z_{\text{norm}} = 1.0) \quad (5)$$

The $z_gradient$ represents the front-minus-back temperature difference through the glass. The fit also provides $z_gradient_err$, the 1σ uncertainty on the through-glass gradient.

8 Surface–Air Coupling: Pre-Control

Before the EAS activates the glycol setpoint, the mirror temperature is governed solely by convective coupling with the cell air (ESS 113). This pre-control period—typically several hours before sunset—provides a clean measurement of the air–mirror thermal relationship at each depth level.

8.1 Pre-Control Offsets

Figure 7 (top right) shows the distribution of $T_{\text{surface}} - \text{ESS 113}$ for each surface during the pre-control period. All three surfaces are close to ESS 113, with small systematic offsets:

- T_{front} is closest to ESS 113 (the optical surface is in direct convective contact with cell air).
- T_{back} tends to be slightly cooler (residual glycol cooling from the previous night).

The top-left panel shows T_{front} vs ESS 113 colored by the through-glass gradient ($z_gradient$), confirming tight coupling between the optical surface and cell air when the glycol is off. Per-night median offsets (bottom-left boxplot) show the inter-night variability, and the bottom-right panel shows an example pre-control night with all three surfaces tracking ESS 113 closely.

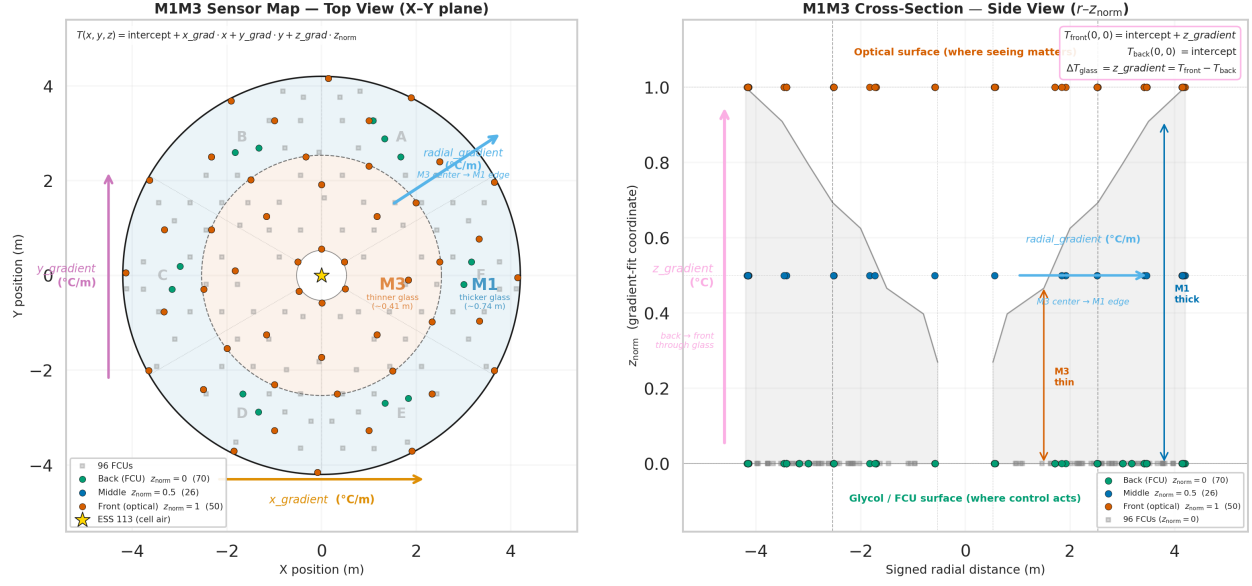


Figure 6: M1M3 sensor layout. Left: top view showing 146 thermocouples (colored by depth level: Front=red, Middle=blue, Back=green), 96 FCUs (gray squares), and ESS 113 (gold star). Arrows indicate the gradient-fit directions. Right: side cross-section with z_{norm} axis (0=back, 1=front), showing the thinner M3 and thicker M1 regions.

9 Surface–Air Coupling: Active Control

Once the glycol activates, the setpoint drives the back surface away from air equilibrium, creating a through-glass gradient. Figure 8 characterizes this behavior.

9.1 Surface Offset Evolution

The top-right panel of Figure 8 shows the median (with IQR band) of $T_{surface} - \text{ESS 113}$ as a function of time since setpoint activation. Key observations:

- All surfaces are driven below ESS 113 by the glycol cooling, consistent with the -1.0°C cold bias.
- T_{back} responds first and most strongly (closest to the FCUs).
- T_{front} lags and has a smaller offset, as the thermal gradient must propagate through the glass.
- The offsets stabilize after ~ 3 – 4 h, consistent with the $\tau \approx 3$ h time constant.

9.2 Through-Glass Gradient vs Setpoint Rate

The bottom-left panel shows $z_gradient$ vs the setpoint rate of change. When the setpoint is decreasing rapidly (aggressive cooling), $z_gradient$ tends negative—the back surface cools faster than the front, as expected from the FCU proximity. When the setpoint is stable, $z_gradient$ clusters near zero.

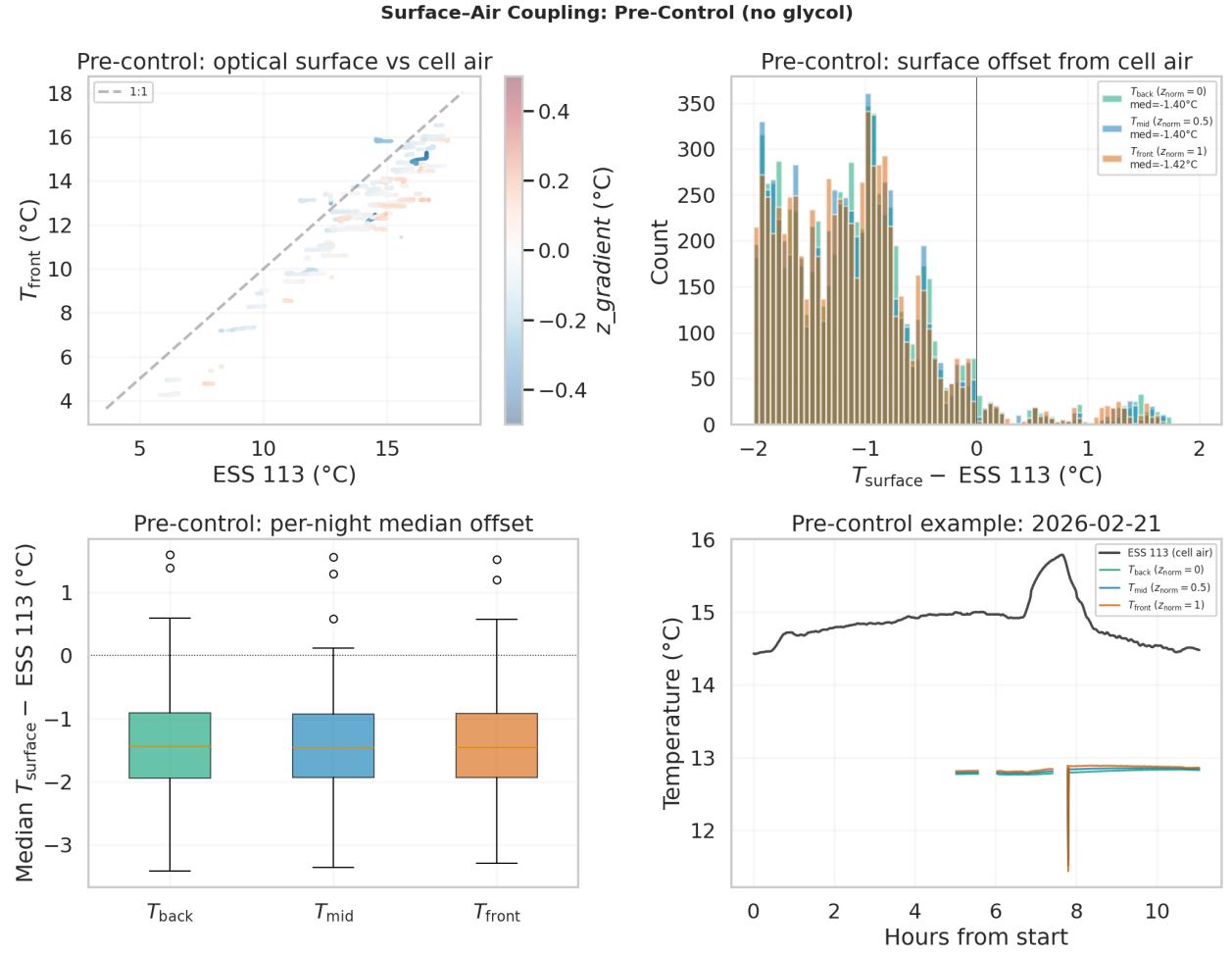


Figure 7: Surface-air coupling during pre-control (no glycol). Top left: T_{front} vs ESS 113 scatter colored by z_{gradient} . Top right: offset distributions for all three surfaces. Bottom left: per-night median offset boxplots. Bottom right: example night showing T_{back} , T_{mid} , T_{front} , and ESS 113.

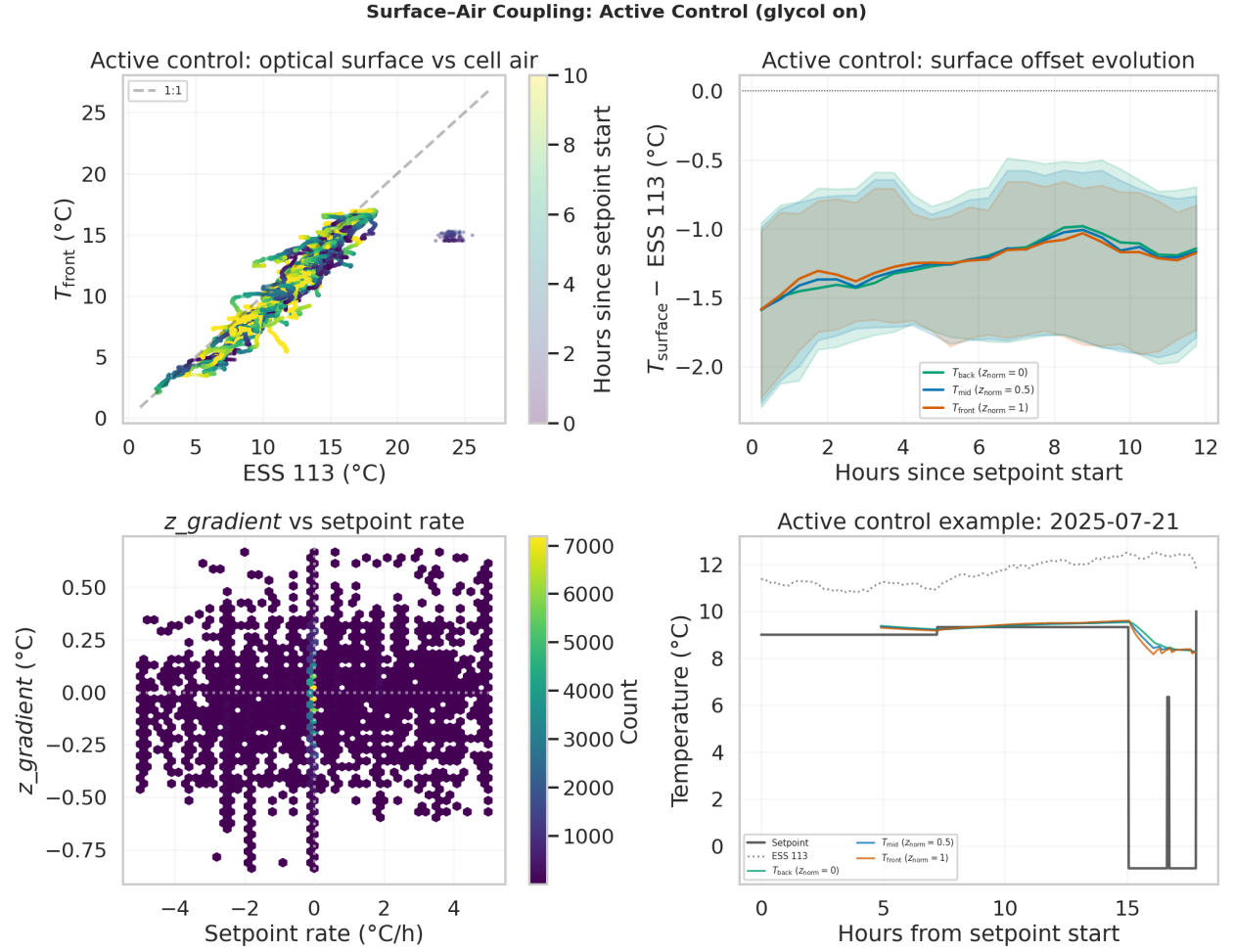


Figure 8: Surface-air coupling during active control. Top left: T_{front} vs ESS 113 colored by hours since activation. Top right: surface offset evolution (median $\pm$ IQR across nights). Bottom left: z_{gradient} vs setpoint rate (hexbin). Bottom right: example night with all surfaces and setpoint.

10 Glycol Response by Surface Depth

The glycol system acts on the back surface through the FCUs. The thermal signal must propagate through the glass to reach the optical surface. We characterize this depth-dependent response using per-surface cross-correlation lag and time constant fits.

10.1 Cross-Correlation Lag

Figure 9 (top left) shows the distribution of cross-correlation lag between the setpoint rate and the rate of change of each surface temperature. The back surface responds first (shortest lag), and the front surface last, as expected from the thermal diffusion through the glass.

10.2 Time Constant by Depth

We fit the glycol-only first-order model $dT_{\text{surface}}/dt = (T_{\text{sp}} - T_{\text{surface}})/\tau$ independently for each surface level. The results (Figure 9, top right) show:

Table 5: Per-surface time constant from glycol-only first-order fit.

Surface	τ median (h)	τ mean (h)
$T_{\text{back}} (z_{\text{norm}} = 0)$	3.0	3.8
$T_{\text{mid}} (z_{\text{norm}} = 0.5)$	2.5	3.4
$T_{\text{front}} (z_{\text{norm}} = 1.0)$	2.0	2.9

The counterintuitive result that $\tau_{\text{front}} < \tau_{\text{back}}$ reflects that the front surface is also coupled to cell air convection (ESS 113), which acts as a second thermal input. The glycol-only model absorbs this additional coupling into a shorter effective τ . The back surface, shielded from air convection by the glass itself, is governed almost entirely by the glycol time constant.

10.3 Glycol Forcing and Through-Glass Gradient

The bottom panels of Figure 9 show the relationship between glycol forcing and the through-glass gradient:

- Bottom left: $z_gradient$ vs (setpoint $- T_{\text{back}}$). When the glycol is actively cooling (setpoint $\ll T_{\text{back}}$), the back cools faster than the front, creating a positive $z_gradient$ (front warmer than back).
- Bottom right: $z_gradient$ vs coolant supply–return ΔT . Strong cooling (large negative $\Delta T_{\text{coolant}}$) correlates with positive $z_gradient$.

11 Gradient Overview: Example Nights

Figure 10 shows four example nights selected by the magnitude of the through-glass gradient $|z_gradient|$: largest, smallest, median, and 25th percentile. Each panel shows the setpoint, ESS 113, and the three surface temperatures (T_{back} , T_{mid} , T_{front}) with the $\pm z_gradient_err$ uncertainty band shaded on T_{front} .

Key observations:

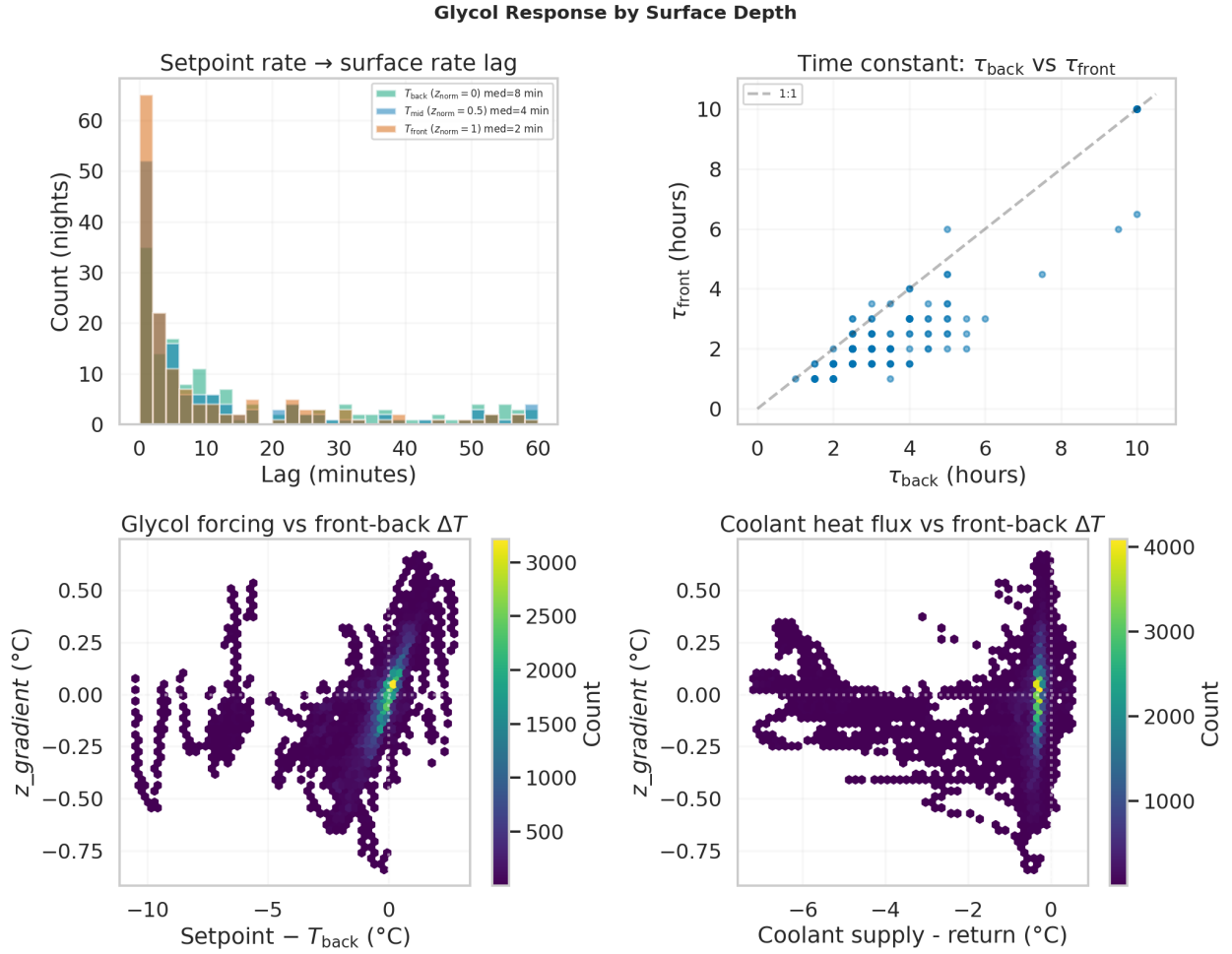


Figure 9: Glycol response by surface depth. Top left: cross-correlation lag (setpoint rate $\rightarrow$ surface rate) per depth level. Top right: τ_{back} vs τ_{front} scatter per night. Bottom left: z_{gradient} vs glycol forcing (setpoint - T_{back}). Bottom right: z_{gradient} vs coolant heat flux proxy (supply - return ΔT).

- On nights with large $|z_gradient|$, the front and back surfaces separate by $\sim 0.3\text{--}0.5^\circ\text{C}$ during active cooling, with the front remaining warmer.
- On nights with small $|z_gradient|$, all three surfaces track each other closely, indicating near-isothermal conditions through the glass.
- The $z_gradient_err$ band is typically $\pm 0.1\text{--}0.2^\circ\text{C}$, confirming that the gradient decomposition is well-constrained by the 96 thermocouples.

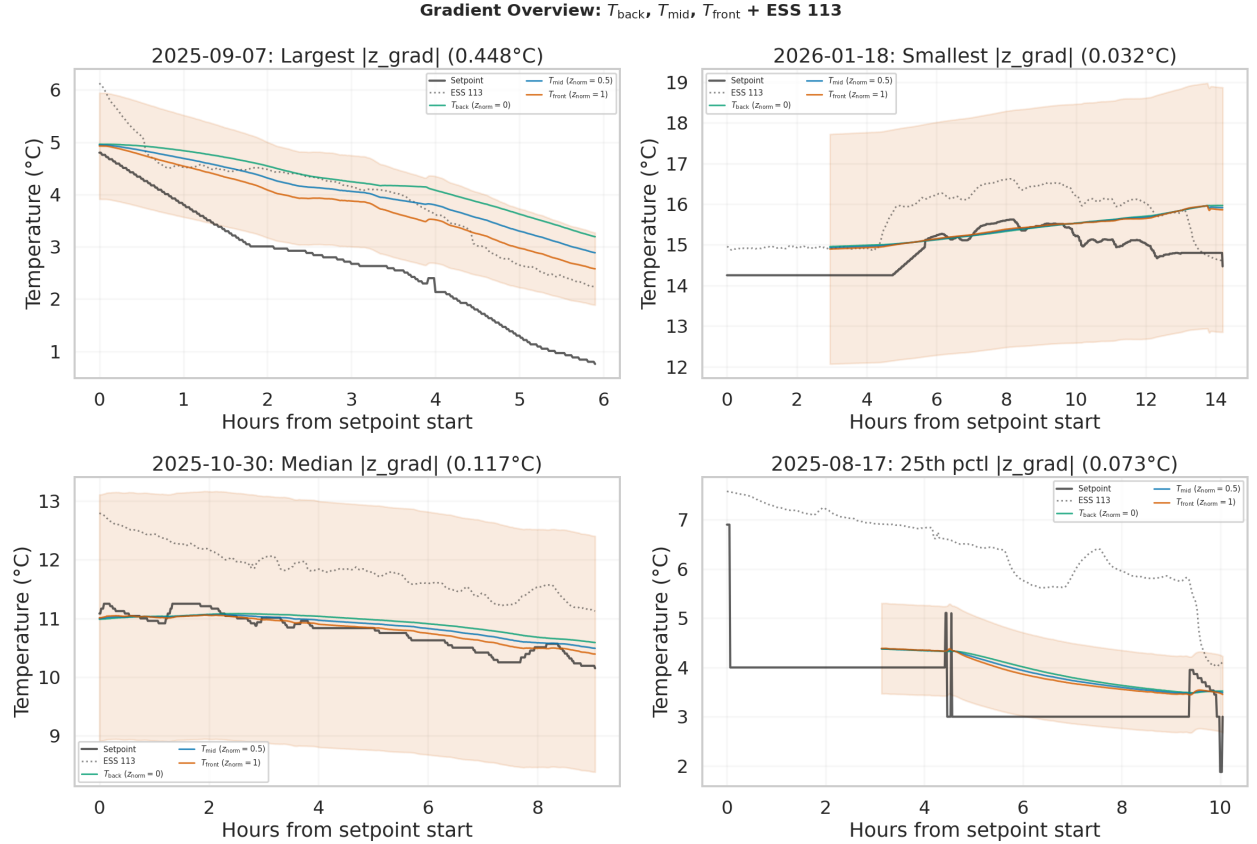


Figure 10: Gradient overview: four example nights selected by $|z_gradient|$ magnitude. Each panel shows setpoint (black), ESS 113 (dotted), and T_{back} (green), T_{mid} (blue), T_{front} (red) with $\pm z_gradient_err$ shaded band.

12 Setpoint–Surface Coupling: Inner Ring vs Outer Ring

The gradient-fit decomposition (Section 10) captures the average through-glass behavior, but the 96 thermocouples span a wide range of radii—from the thin M3 inner region ($r \approx 0.56\text{ m}$, glass thickness 0.24 m) to the thick M1 outer region ($r > 3.5\text{ m}$, glass thickness 0.74 m). To resolve the radial dependence of the glycol coupling, we use **raw thermocouple readings** (not the gradient-fit decomposition) averaged over two concentric rings at two depth levels.

12.1 Data and Method

The inner ring uses 14 MTCIW sensors at $r \approx 0.56$ m (6 front, 2 middle, 6 back), averaged into `inner_front` and `inner_back`. The outer ring uses all sensors at $r > 3.5$ m, similarly averaged. These are merged with the ESS 113, setpoint, and dome status from the thermal parquet (July 2025+) and a supplementary ESS 113 query (May–June 2025), yielding 155 valid nights with concurrent setpoint and thermocouple data.

For each night and each of the four surfaces (inner front, inner back, outer front, outer back), we:

1. Fit the first-order model $dT/dt = (T_{\text{sp}} - T)/\tau$ by grid search over $\tau \in [0.5, 12]$ h in 0.25 h steps.
2. Compute the cross-correlation lag between the setpoint rate and the surface temperature rate.

12.2 Time Constant by Ring and Depth

Table 6: Setpoint-to-surface thermal coupling: raw thermocouple results (155 nights, cleaned setpoint—gaps > 5 min excluded).

Surface	τ median (h)	Lag median (min)	RMSE median ($^{\circ}\text{C}$)
Inner front ($r \approx 0.56$ m, $z = 1$)	2.00	4	0.098
Inner back ($r \approx 0.56$ m, $z = 0$)	3.00	9	0.066
Outer front ($r > 3.5$ m, $z = 1$)	2.00	3	0.105
Outer back ($r > 3.5$ m, $z = 0$)	3.00	6	0.070

The results (Table 6, Figure 11) reveal clear physical trends:

- **Front surfaces respond faster than back surfaces** at both rings (inner: 2.0 vs 3.0 h; outer: 2.0 vs 3.0 h). This confirms the gradient-fit finding (Section 10) and reflects the dual coupling of the front surface to both glycol and cell air convection.
- **Back surfaces converge at $\tau = 3.0$ h** regardless of ring radius, suggesting that the glycol-to-back coupling is dominated by the FCU contact resistance rather than glass thickness.
- **The back surfaces are the thermal bottleneck:** $\tau = 3.0$ h at both rings, with the outer back having the highest RMSE (0.070 $^{\circ}\text{C}$) and longest lag (6 min).
- **Front surfaces respond in ~ 2 h** at both rings, with lags of only 3–4 min, due to the additional convective coupling with cell air.
- **Inner back has the lowest RMSE** (0.066 $^{\circ}\text{C}$), confirming that the thin-glass back surface most directly reflects the glycol setpoint.

12.3 Through-Glass Gradient by Ring

The through-glass temperature difference (front $-$ back) is near zero on average at both rings (inner median: +0.002 $^{\circ}\text{C}$, outer median: -0.006 $^{\circ}\text{C}$), but the inner ring shows larger scatter ($\sigma = 0.18$ $^{\circ}\text{C}$ vs 0.15 $^{\circ}\text{C}$ for the outer ring).

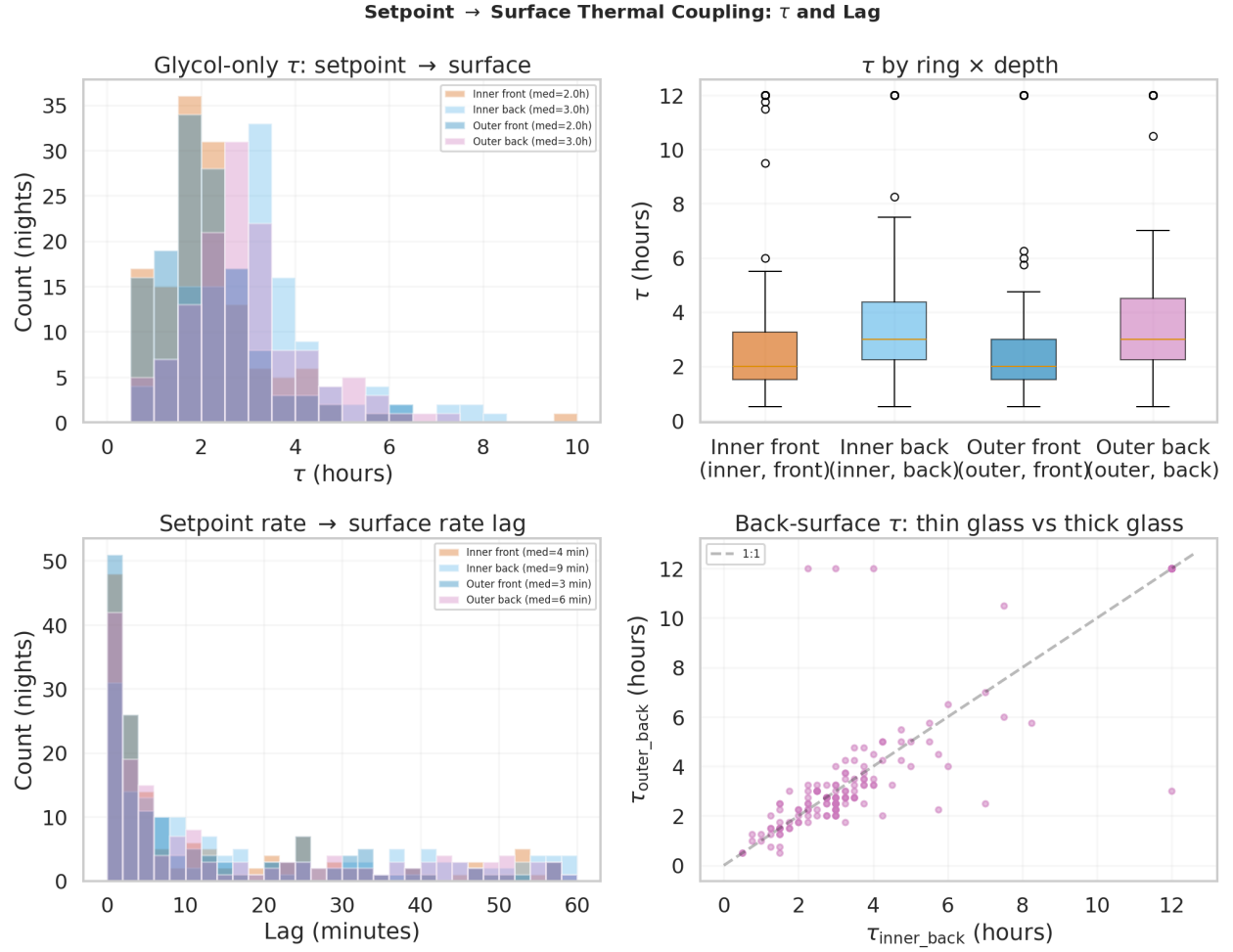


Figure 11: Setpoint-to-surface coupling: τ and lag. Top left: τ distributions by surface. Top right: τ boxplots by ring $\times$ depth. Bottom left: cross-correlation lag distributions. Bottom right: $\tau_{\text{inner_back}}$ vs $\tau_{\text{outer_back}}$ (same depth, different ring thickness).

Figure 12 (top right, bottom left) shows that when the glycol is strongly cooling (setpoint $\ll T_{\text{back}}$), the inner ring develops a clear negative front-back gradient (back cooling faster than front), reaching $\pm 0.5^\circ\text{C}$. The outer ring shows a tighter gradient range ($\pm 0.3^\circ\text{C}$), as the thicker glass dampens the through-thickness temperature difference.

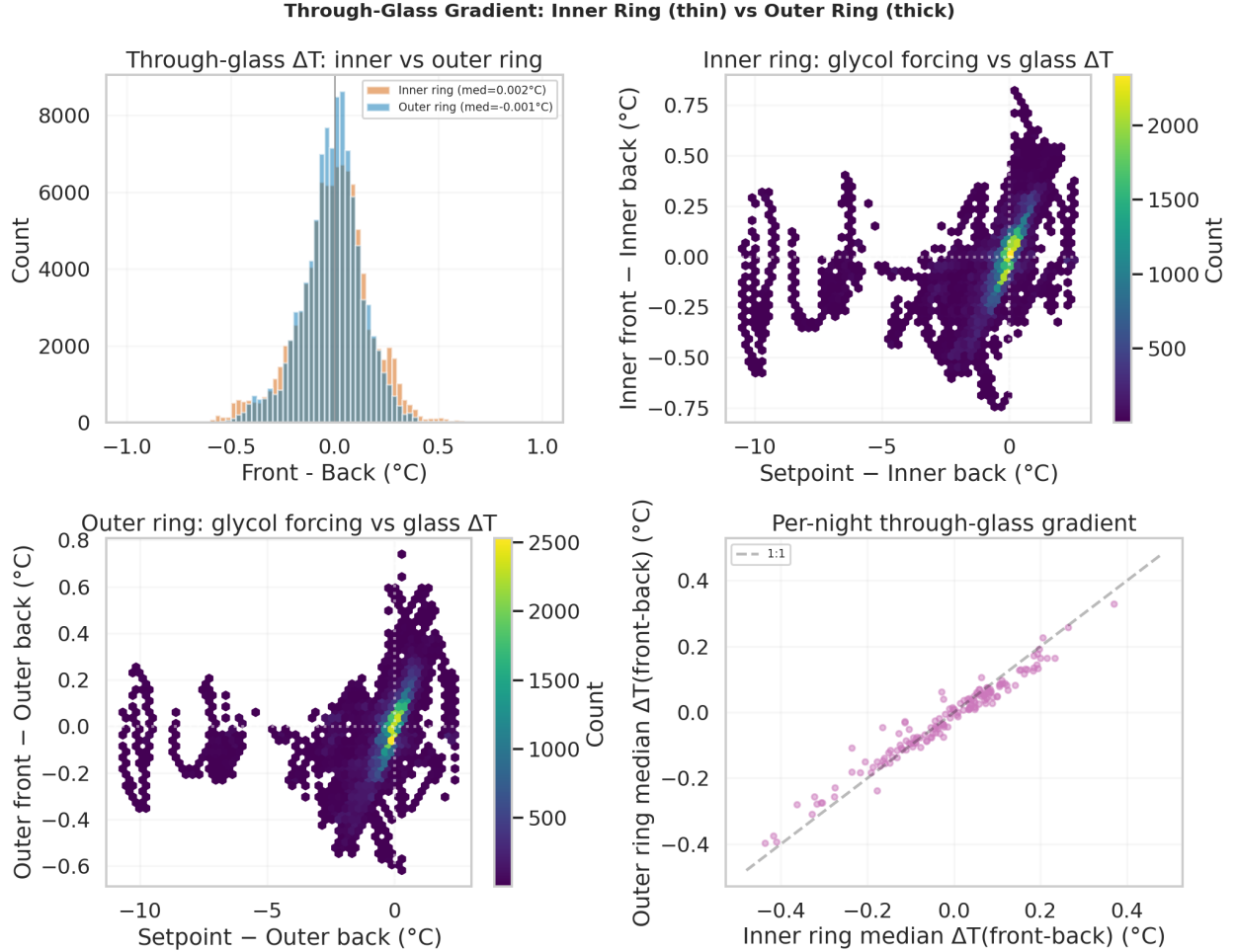


Figure 12: Through-glass gradient: inner ring (thin, 0.24 m) vs outer ring (thick, 0.74 m). Top left: front-back ΔT distributions. Top right: inner ring glass ΔT vs glycol forcing (hexbin). Bottom left: outer ring glass ΔT vs glycol forcing. Bottom right: per-night median inner vs outer gradient.

12.4 Gradient-Forcing Slope: Safe Setpoint Range

To determine the glycol forcing that can be applied without creating excessive through-glass gradients, we fit the relationship

$$\Delta T_{\text{glass}} = T_{\text{front}} - T_{\text{back}} = \alpha \cdot (T_{\text{sp}} - T_{\text{back}}) + \beta \quad (6)$$

using Theil-Sen robust regression (resistant to outliers) on 80,361 minute-level samples per ring (Figure 13).

The slope α gives the fraction of the glycol forcing that appears as a through-glass gradient. All three measures are consistent: $\alpha \approx 0.18\text{--}0.22^\circ\text{C}/^\circ\text{C}$. The 96-thermocouple z_{gradient} (which

Table 7: Theil-Sen robust fit: through-glass gradient vs glycol forcing.

Gradient measure	Slope α ($^{\circ}\text{C}/^{\circ}\text{C}$)	Intercept β ($^{\circ}\text{C}$)	N samples
$z_gradient$ (96-TC fit)	0.223	-0.011	86,472
Inner ring (0.24 m glass)	0.212	$+0.006$	80,361
Outer ring (0.74 m glass)	0.181	$+0.004$	80,361

averages over all radii) has the steepest slope (0.223), while the inner ring raw TCs give 0.212 and the outer ring gives 0.181—the thinner inner glass transmits more of the forcing as a gradient.

Using these slopes, we can infer the safe forcing range for a given gradient threshold:

Table 8: Maximum glycol forcing ($T_{\text{sp}} - T_{\text{back}}$) to stay within gradient limits.

$ \Delta T_{\text{glass}} $ limit	$z_gradient$ forcing	Inner ring forcing	Outer ring forcing
0.1°C	$\pm 0.4^{\circ}\text{C}$	$\pm 0.5^{\circ}\text{C}$	$\pm 0.6^{\circ}\text{C}$
0.2°C	$\pm 0.8^{\circ}\text{C}$	$\pm 1.0^{\circ}\text{C}$	$\pm 1.1^{\circ}\text{C}$
0.3°C	$\pm 1.3^{\circ}\text{C}$	$\pm 1.4^{\circ}\text{C}$	$\pm 1.7^{\circ}\text{C}$

For the typical EAS cold bias of -1.0°C , the expected glass gradient is $\sim 0.2^{\circ}\text{C}$ (all measures)—well within safe limits. Aggressive transient forcing of -5°C (as seen during rapid cool-downs) would produce $\sim 1^{\circ}\text{C}$ gradients at the inner ring.

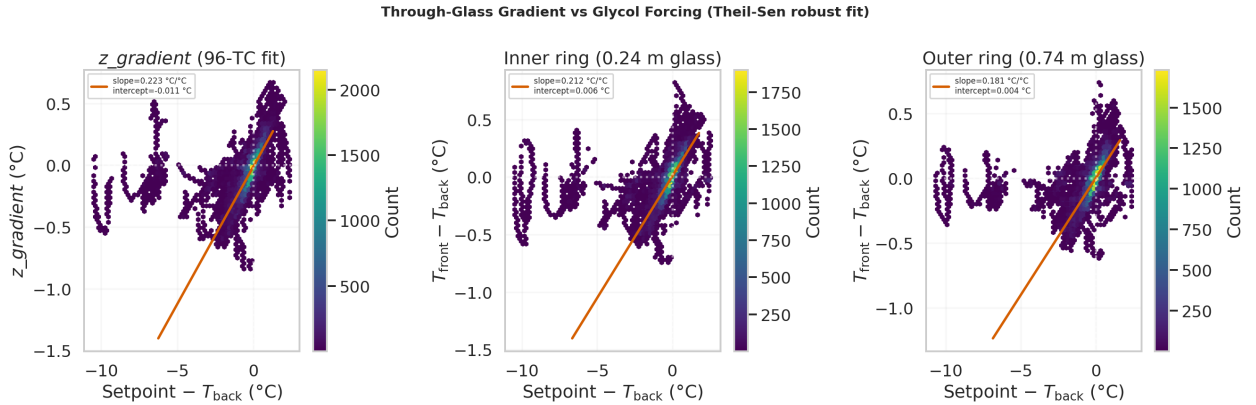


Figure 13: Through-glass gradient vs glycol forcing ($T_{\text{sp}} - T_{\text{back}}$) with Theil-Sen robust linear fit (red line). Left: 96-TC gradient-fit $z_gradient$ (slope = 0.223). Center: inner ring raw TCs (0.24 m glass, slope = 0.212). Right: outer ring raw TCs (0.74 m glass, slope = 0.181).

12.5 Glycol Coolant: Active vs Passive Operation

The glycol coolant loop has 97% time coverage, while setpoint commands cover only 8.9% of the data. The coolant circulates continuously, but does its heat flux actually drive the mirror temperature?

12.5.1 Coolant ΔT Does Not Predict Mirror Cooling Rate

We test the causal relationship by correlating the coolant heat flux proxy ($\Delta T_{\text{coolant}} = T_{\text{supply}} - T_{\text{return}}$) with the 5-minute smoothed mirror cooling rate dT_{mirror}/dt (Figure 14).

Table 9: Pearson correlation r between mirror cooling rate (dT/dt inner back) and candidate drivers.

Driver	Period	r
Setpoint $- T_{\text{back}}$	Active	0.577
Coolant ΔT (supply $-$ return)	Active	0.075
Coolant ΔT (supply $-$ return)	Passive	0.024

The results (Table 9) are unambiguous:

- The **setpoint error strongly predicts** the mirror cooling rate ($r = 0.577$, Figure 14 bottom left). When the setpoint is far below the mirror, the mirror cools at up to $-1.5^\circ\text{C}/\text{h}$; near equilibrium, $dT/dt \approx 0$. The Theil-Sen slope is $0.24 (^\circ\text{C}/\text{h})/^\circ\text{C}$, consistent with $\tau \approx 1/0.24 \approx 4\text{h}$.
- The **coolant ΔT does not predict** the mirror cooling rate—neither during active ($r = 0.075$) nor passive ($r = 0.024$) operation. Despite coolant ΔT reaching -6°C during active cooling, it shows essentially no linear correlation with dT_{mirror}/dt .

12.5.2 Interpretation

The coolant ΔT reflects how much heat the glycol *has already absorbed* from the mirror, not how much cooling it will *deliver next*. The 96 FCUs modulate their local flow and fan speed based on the commanded setpoint—the setpoint governs *how much* of the available chiller capacity is directed at the mirror. The coolant supply–return ΔT is therefore a *consequence* of the setpoint-driven cooling, not its cause.

12.5.3 Energy Balance

Despite the lack of predictive correlation, the energy balance relates the coolant ΔT to the mirror’s thermal state. The heat extraction rate through the glycol loop is:

$$\dot{Q} = \dot{m} c_p (T_{\text{return}} - T_{\text{supply}}) \quad (7)$$

where $\dot{m}$ is the mass flow rate and c_p the specific heat. For 50% propylene glycol at $\sim 10^\circ\text{C}$: $\rho = 1040 \text{ kg}/\text{m}^3$, $c_p = 3560 \text{ J}/(\text{kg K})$.

Equating coolant heat extraction to the mirror’s temperature change:

$$\dot{m} c_p |\Delta T_{\text{coolant}}| = M_{\text{mirror}} c_{\text{glass}} \left| \frac{dT_{\text{mirror}}}{dt} \right| \quad (8)$$

For M1M3: $M_{\text{mirror}} \approx 30,000 \text{ kg}$, $c_{\text{glass}} \approx 670 \text{ J}/(\text{kg K})$ (borosilicate). From $\tau = 3 \text{ h}$, the effective thermal conductance is:

$$h_{\text{eff}} A = \frac{M_{\text{mirror}} c_{\text{glass}}}{\tau} = \frac{30,000 \times 670}{10,800} \approx 1860 \text{ W/K} \quad (9)$$

At a typical cooling rate of $-1.0^\circ\text{C}/\text{h}$, the required heat extraction is $\dot{Q} \approx 5.6 \text{ kW}$. With $\Delta T_{\text{coolant}} = -4^\circ\text{C}$, this implies $\dot{m} \approx 0.4 \text{ kg/s}$ ($\sim 6 \text{ GPM}$ across 96 FCUs).

12.5.4 Passive Circulation

During passive operation (no setpoint, 91% of data), $\Delta T_{\text{coolant}} \approx 0^\circ\text{C}$ (Figure 14, top left). The glycol circulates to maintain hydraulic readiness but the FCUs are throttled to minimum flow, extracting negligible heat. The mirror temperature during these periods drifts primarily in response to ambient air convection, not the coolant system ($r = 0.024$).

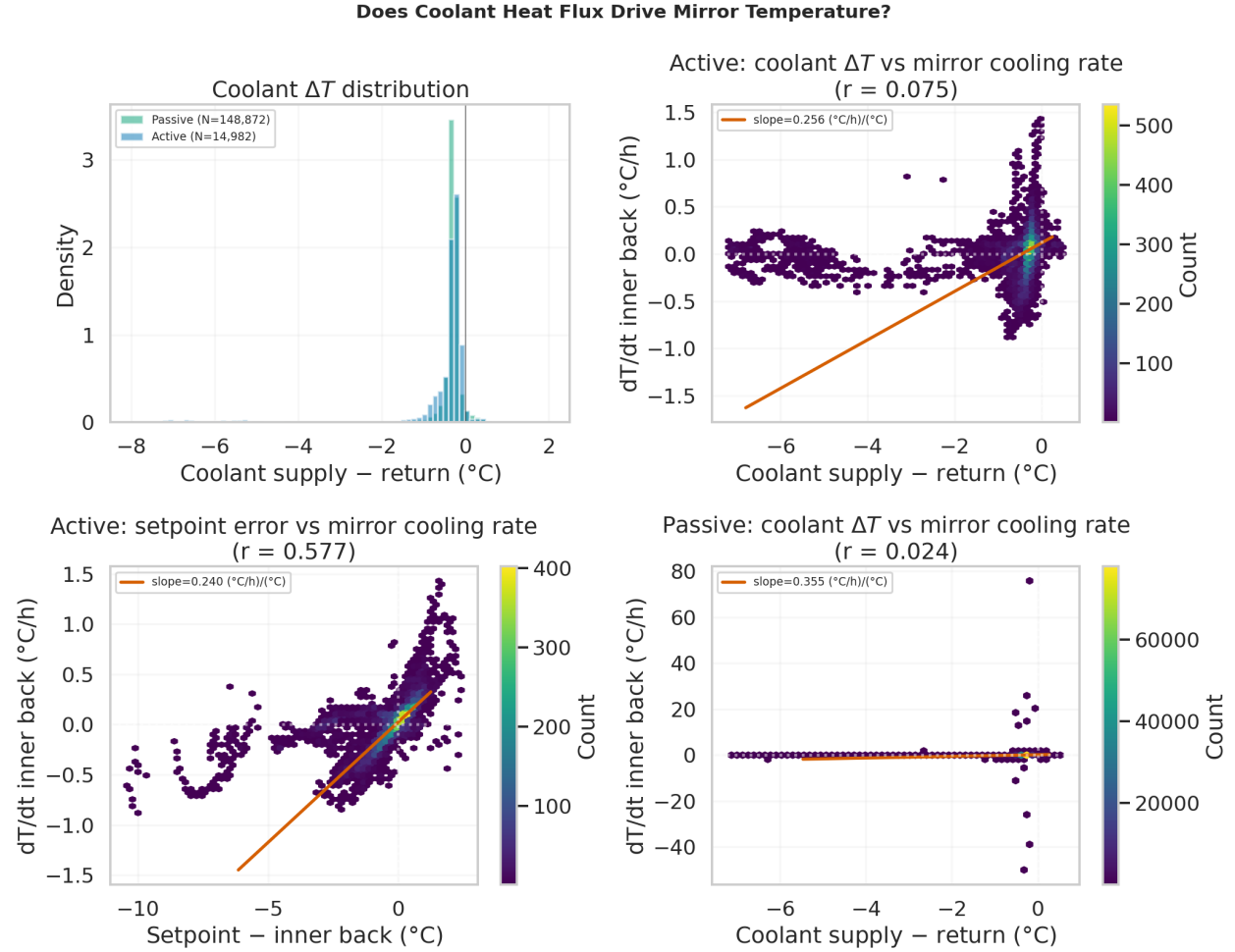


Figure 14: Does coolant heat flux drive the mirror? Top left: coolant ΔT distribution—passive (green, near zero) vs active (blue, reaching -6°C). Top right: dT/dt vs coolant ΔT during active control ($r = 0.075$, no correlation). Bottom left: dT/dt vs setpoint error ($r = 0.577$, strong correlation). Bottom right: dT/dt vs coolant ΔT during passive circulation ($r = 0.024$, no correlation). The setpoint error is the driver; the coolant ΔT is a consequence.

12.6 Example Nights and Model Quality

Figure 15 shows four example nights: fastest and slowest outer-ring τ , best and median model fit. The first-order model (dashed black) captures the thermal response well for most nights. On nights with large τ (top right), the surfaces barely respond to the setpoint—these are typically nights where the setpoint changes are small and the mirror is already near equilibrium.

Figure 16 summarizes the model fit quality. The RMSE distributions (top left) cluster below $0.25\text{ }^{\circ}\text{C}$ for all four surfaces, confirming that the first-order model captures the dominant physics even at individual thermocouple locations. The $\tau_{\text{inner_front}}$ vs $\tau_{\text{outer_front}}$ scatter (bottom left) shows that the outer optical surface consistently requires a longer time constant, reflecting the thicker glass path.

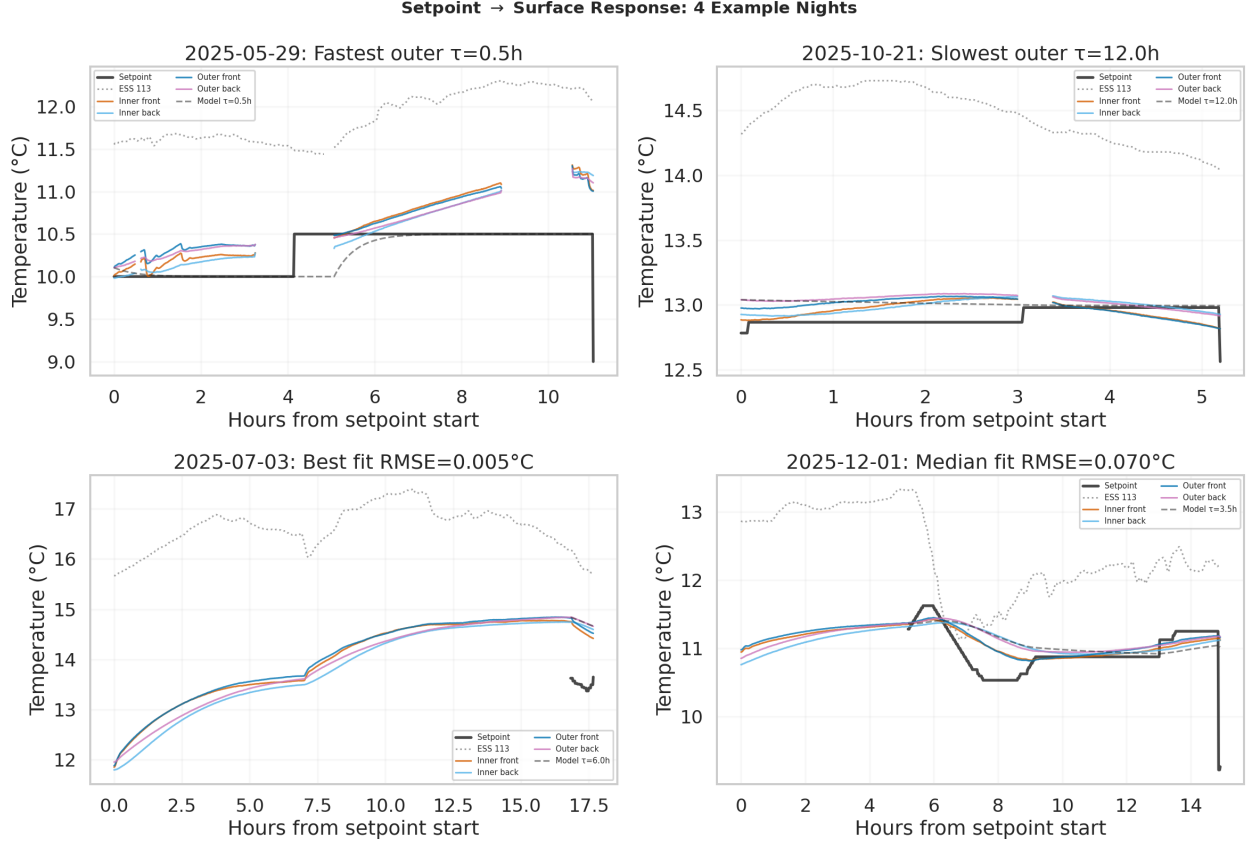


Figure 15: Four example nights showing setpoint (black), ESS 113 (dotted), and all four surfaces (inner front/back, outer front/back). The dashed line shows the first-order model fit for the outer back surface. Panels selected by: fastest τ (top left), slowest τ (top right), best RMSE (bottom left), median RMSE (bottom right).

13 Implications for Control Simulation

1. **Cold bias should be $1.0\text{ }^{\circ}\text{C}$** , not $0.3\text{ }^{\circ}\text{C}$. The EAS targets ESS 113 – $1.0\text{ }^{\circ}\text{C}$.
2. **Bulk time constant is $\tau = 2.5\text{ h}$** (for T_{mid}). Surface-dependent: $\tau_{\text{back}} = 3.0\text{ h}$, $\tau_{\text{front}} = 2.0\text{ h}$.
3. **Rate limit of $0.7\text{ }^{\circ}\text{C/h}$ is conservative.** The EAS routinely exceeds this ($p95 = 1.8\text{ }^{\circ}\text{C/h}$), confirming that the soft-rate-penalty approach (allowing occasional violations up to $\sim 2\text{ }^{\circ}\text{C/h}$) is physically realistic.
4. **The first-order model is excellent for the bulk mirror** when validated against T_{mid} ($z_{\text{norm}} = 0.5$).

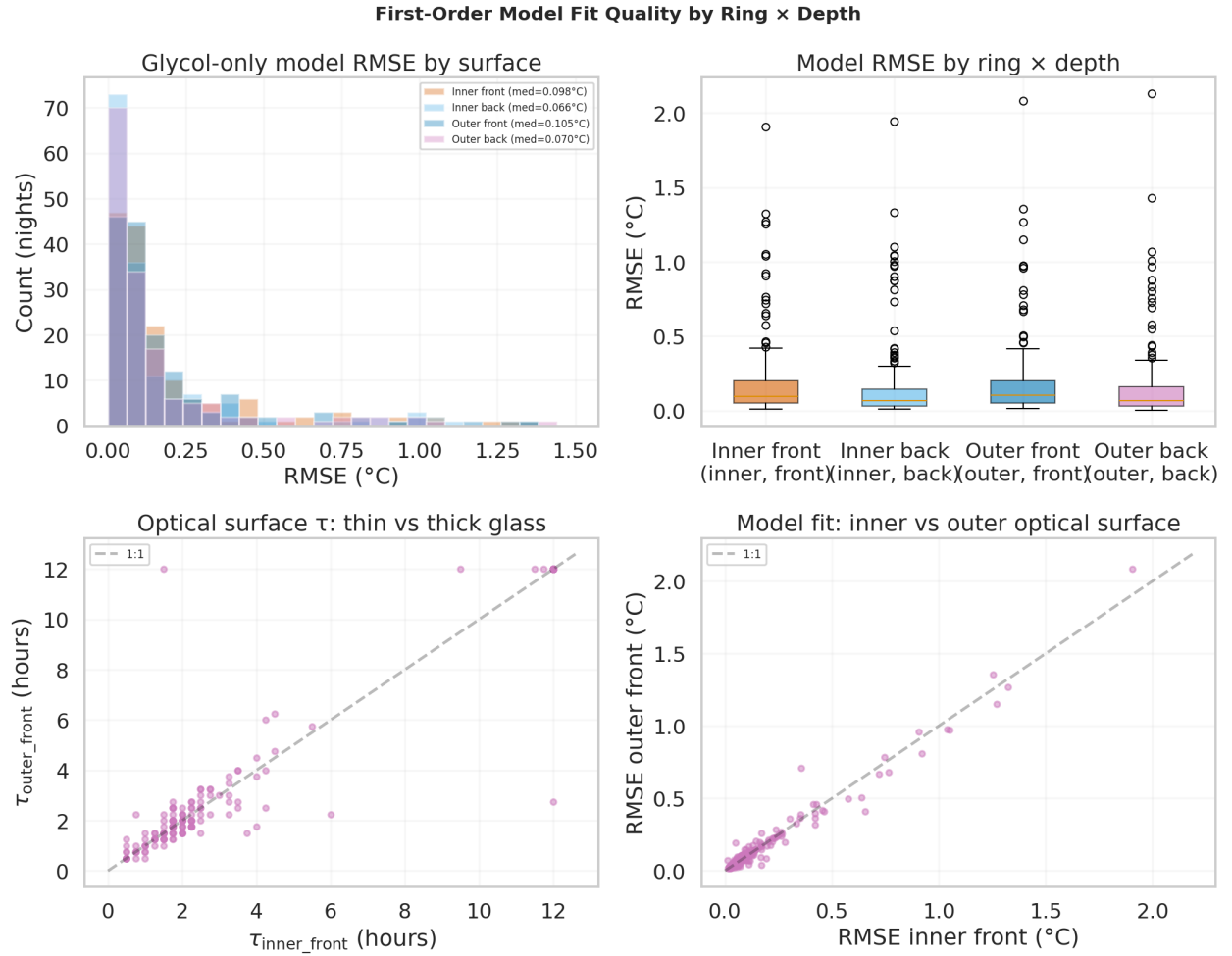


Figure 16: Model fit quality by ring $\times$ depth. Top left: RMSE distributions. Top right: RMSE boxplots. Bottom left: $\tau_{\text{inner_front}}$ vs $\tau_{\text{outer_front}}$ scatter. Bottom right: RMSE inner front vs RMSE outer front.

5. **Current EAS achieves 26.5% within $\pm 0.3^\circ\text{C}$.** This is the baseline to beat with improved forecasting and control.
6. **The optical surface (T_{front}) is coupled to both glycol and cell air.** Its effective $\tau_{\text{front}} = 2.0\text{ h}$ is shorter than $\tau_{\text{back}} = 3.0\text{ h}$ because convective coupling with ESS 113 provides a second thermal input. A two-input model may better represent the front-surface dynamics.
7. **The through-glass gradient z_{gradient} is a diagnostic for control aggressiveness.** Large $|z_{\text{gradient}}|$ ($> 0.3^\circ\text{C}$) indicates the glycol is driving the back surface faster than the front can follow. Future controllers could monitor z_{gradient} as a constraint to limit thermal stress.
8. **Back surfaces are the thermal bottleneck** ($\tau = 3.0\text{ h}$ at both rings), while front surfaces respond in $\sim 2\text{ h}$ due to additional air convection coupling. The inner back has the lowest RMSE (0.070°C), confirming it most directly reflects the glycol setpoint.
9. **Glass thickness affects the front surface more than the back.** Back-surface τ is the same at both rings (3.0 h), suggesting FCU contact resistance dominates over glass conduction for the glycol-facing surface.

14 Conclusions

1. The EAS targets $T_{\text{setpoint}} = \text{ESS 113} - 1.0^\circ\text{C}$, updating every 3 min at rates up to 1.8°C/h (p95).
2. The glycol system responds to setpoint changes with a 4-minute lag (at T_{mid}). The bulk glass temperature tracks the setpoint within -0.07°C (median offset).
3. The first-order thermal model with $\tau = 2.5\text{ h}$ achieves median $\text{RMSE} = 0.059^\circ\text{C}$ against T_{mid} —an excellent fit that validates the model structure.
4. The bulk glass time constant is $\tau \approx 2.5\text{ h}$ (T_{mid}), with depth-dependent values from $\tau_{\text{back}} = 3.0\text{ h}$ to $\tau_{\text{front}} = 2.0\text{ h}$.
5. The current EAS achieves per-night $\text{RMS} = 0.74^\circ\text{C}$ and 26.5% within $\pm 0.3^\circ\text{C}$ —significant room for improvement with better forecasting and control algorithms.
6. The gradient-fit decomposition resolves three depth levels through the glass. The effective time constant decreases from $\tau_{\text{back}} = 3.0\text{ h}$ to $\tau_{\text{front}} = 2.0\text{ h}$, reflecting the dual coupling of the optical surface to both glycol and cell air convection.
7. The through-glass gradient z_{gradient} is typically $< 0.3^\circ\text{C}$ but can reach 0.5°C during aggressive cooling. The gradient uncertainty ($z_{\text{gradient_err}} \approx 0.1\text{--}0.2^\circ\text{C}$) confirms the decomposition is well-constrained.
8. Pre-control (no glycol), all three surfaces track ESS 113 closely with small offsets, confirming good air–mirror thermal equilibrium before sunset.
9. Raw thermocouple analysis of inner ring ($r \approx 0.56\text{ m}$, glass thickness 0.24 m) vs outer ring ($r > 3.5\text{ m}$, 0.74 m) over 155 nights (with cleaned setpoint signal) shows: back surfaces converge at $\tau = 3.0\text{ h}$ regardless of radius (FCU contact resistance dominates), while front surfaces respond in $\sim 2\text{ h}$ due to air convection coupling. Model RMSE is $0.07\text{--}0.10^\circ\text{C}$ across all four ring $\times$ depth combinations.

10. Theil-Sen robust regression of the through-glass gradient vs glycol forcing yields slopes of $0.21\text{ }^{\circ}\text{C}/^{\circ}\text{C}$ (inner) and $0.18\text{ }^{\circ}\text{C}/^{\circ}\text{C}$ (outer). This means the EAS's typical $-1\text{ }^{\circ}\text{C}$ forcing produces only $\sim 0.2\text{ }^{\circ}\text{C}$ glass gradient. To keep gradients below $0.3\text{ }^{\circ}\text{C}$, forcing should stay within $\pm 1.4\text{ }^{\circ}\text{C}$ (inner) or $\pm 1.7\text{ }^{\circ}\text{C}$ (outer).